

100 Days of Cognitive Augmentation Challenge

Day 3

QUANTIFYING DISTRACTION

Modeling Bottom-Up Saliency in Physical Environments.

Keywords: Attention Economy, Computer Vision, Evolutionary Mismatch.

Evolutionary Mismatch Theory.

Dual-Process Theory:

Attention is a competition between **Top-Down** (Goal-Directed, Prefrontal Cortex) and **Bottom-Up** (Stimulus-Driven, Primal).

The Evolutionary Proxy:

Our visual system is optimized for high contrast, intensity, and sudden motion – markers that historically signaled survival threats.

The Mismatch:

Modern design exploits these "Saliency features." We are not dealing with a failure of discipline; we are dealing with a **biological compromise** against an adversarial environment.

Spectral Residual Approach.

The Stack:

Python environment leveraging OpenCV for real-time video processing.

The Mechanism:

1. **Fourier Transform:** We analyze the image in the frequency domain to establish statistical regularity (the "background").
2. **Residual Extraction:** We subtract the averaged log-spectrum to isolate the "innovations"— the features that mathematically deviate from the norm.

The Output:

A probability map where high-frequency noise (clutter, notifications) renders as a **Heatmap (H)**, while predictable patterns fade to black.

Results



Implications & Future Work : Augmentation and Mediated Reality.

Diagnostic Utility: A "Digital Mirror" to identify **Environmental Latency** — visual cues that trigger subconscious processing loads.

Intervention: Identification leads to elimination (e.g., removing high-contrast clutter).

Future Roadmap:

This algorithm is the precursor to **Active Dimming AR**.

Moving from *detecting* distraction to *optically suppressing* it in real-time.

We are building a focus mode for the physical world.